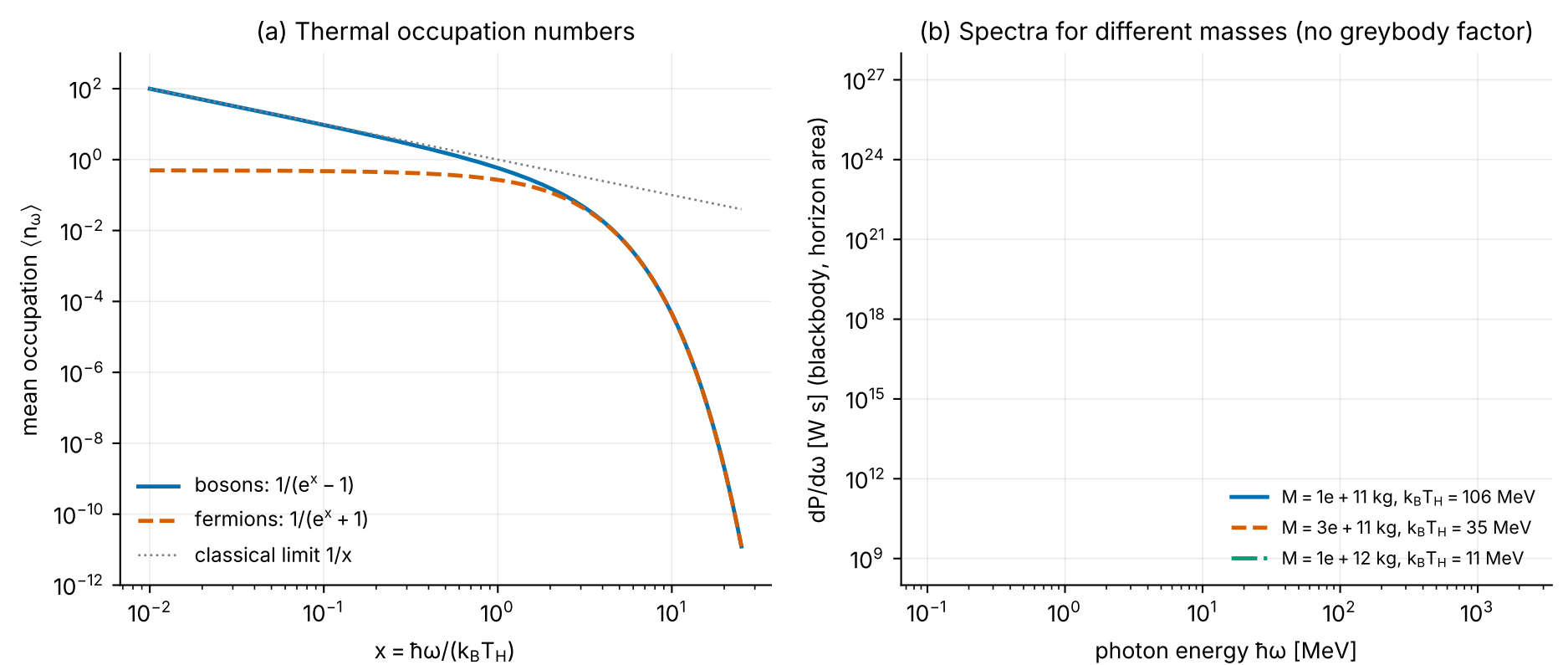
(a) Thermal occupation numbers
mean occupation ⟨n_ω⟩
x = ħω/(k_B T_H)
bosons: 1/(e^x − 1)
fermions: 1/(e^x + 1)
classical limit 1/x
(b) Spectra for different masses (no greybody factor)
dP/dω [W s] (blackbody, horizon area)
photon energy ħω [MeV]
M = 1e + 11 kg, k_B T_H = 106 MeV
M = 3e + 11 kg, k_B T_H = 35 MeV
M = 1e + 12 kg, k_B T_H = 11 MeV